A DIRECT SUM OF SIMPLE LIE ALGEBRAS OVER Q IS R-UNIVERSAL EXACTLY UNDER FOUR BRAUER-THEORETIC CONDITIONS

DAVE WITTE MORRIS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned publisher source sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the source pins the publisher class sigma, which is not part of the delivery baseline (supported wrapper classes are article and amsart), so this card uses sources/algebra-fl02-a-cohomological-proof-that-real-re/retained-preamble.tex. That file supplies the amsart baseline, then copies the author's DeclareMathOperator list, equation numbering and newtheorem declarations from pinned source lines 11-31, and the author's own \Res, \ResOne and \Lie macros from pinned source lines 35-43, which the author writes immediately after \begin{document} in the publisher layout. The publisher front-matter commands of that class and the macro \PaperNumber they require are not reproduced because this excerpt carries neither the journal front matter nor anything that uses them. The two operator names the author declares that the delivery baseline already supplies (\GL and \SL) are left to the baseline, so that no declaration conflicts; their printed meaning is unchanged. No macro was invented and no mathematical symbol was changed.

Editorial note. Bibliography provenance: the counted statement and its proof contain no citation at all, so this card carries no bibliography. The source does have a thebibliography environment (pinned source lines 954-1016) with three entries used elsewhere in the paper; nothing from it is needed here and no entry was reconstructed.

Editorial note. External references are navigation pointers only; every deliberately kept step of the author's argument stays inside the counted spans. Their printed numbers are the author's own theorem numbering under the author's \newtheorem declarations, computed mechanically by the lane's scanner.

Editorial note. The author writes the proposition label as a LaTeX comment on source line 886 (the line reads percent-backslash-label-SS), so the environment is unlabelled in the author file; the proof refers to the four numbered conditions by their item labels (SS-Gi, SS-star, SS-Di=Dj, SS-Dsplit), all of which lie inside the counted statement span and therefore resolve inside the delivered material.

Allowed standard prerequisites (editorial).

- Brauer equivalence of central simple algebras, denoted by the author with the symbol is-Brauer-equivalent-to, and the classification of real division algebras. Used inside the counted proof (source lines 915, 926-932, 938-941); the author introduces the notation in the first line of the proof. Turns the Brauer-equivalence computations for the tensor products D(Lie

Date: doi:10.3842/SIGMA.2015.034; CC BY 4.0.

G, R)(lambda_i) tensor D(Lie G, R)(lambda_j) and for the splits-over-L statements into the isomorphism and splitting conclusions of conditions (3) and (4).

- Tits algebras D(Lie G, K)(lambda) attached to dominant weights and their behaviour under scalar extension. Used inside the counted proof at source lines 926-932 and 938-941; the author sets up in the source preceding the counted statement. Supplies the objects whose Brauer classes are being compared, and the identity D(Lie G, K)(lambda_i + lambda_j) = D(Lie G, L)(lambda_i + lambda_j) over the splitting field L.

Source reference. Original lemma 2.3: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 471–471. Navigation pointer only: the counted proof refers to this lemma in one line for condition (2), and the card declares the content of that step as a named standard prerequisite instead of reproducing the lemma.

Source reference. Original lemma 2.1: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 416–416. Navigation pointer only: the counted proof invokes this lemma once, in the last paragraph for condition (4), for the equivalence between R-universality and the division-algebra condition; that equivalence is declared as a named standard prerequisite.

Source reference. Original definition 4: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 432–432. Navigation pointer only: it is the fourth numbered item of lemma RunivIff, which the counted proof appeals to once in the last paragraph for condition (4); the pointer is here so that the author's own cross-reference number resolves, and the mathematical content it names is declared as a named standard prerequisite.

Proposition 0.1. *A direct sum $\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_r$ of simple Lie algebras over $\mathbb{Q}$ is $\mathbb{R}$-universal if and only if:*

(1) *each $\mathfrak{g}_i$ is $\mathbb{R}$-universal,*
(2) *either $\mathfrak{g}$ is inner, or it becomes inner over an imaginary quadratic extension L of $\mathbb{Q}$, and the $*$-action of $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$ is the same as the $*$-action of $\mathrm{Gal}(L/\mathbb{Q})$.*
(3) *we have $D_{\mathfrak{g}_i,\mathbb{Q}}(\lambda_i) \cong D_{\mathfrak{g}_j,\mathbb{Q}}(\lambda_j)$ whenever λ_i and λ_j are $*$-invariant dominant weights of G_i and G_j, such that $D_{\mathfrak{g}_i,\mathbb{Q}}(\lambda_i) \neq \mathbb{Q}$ and $D_{\mathfrak{g}_j,\mathbb{Q}}(\lambda_j) \neq \mathbb{Q}$, and*
(4) *either $\mathfrak{g}$ is inner, or it becomes inner over an imaginary quadratic extension L of $\mathbb{Q}$, such that $D_{\mathfrak{g}_i,\mathbb{Q}}(\lambda_i)$ splits over L, for every dominant weight λ_i of every $\mathfrak{g}_i$ that is inner (over $\mathbb{Q}$).*

Proof. We prove only ($\Rightarrow$), but the argument is reversible. (We use $\sim$ to denote "is Brauer equivalent to".)

(1) Any representation ρ_i of $\mathfrak{g}_i$ extends to a representation of $\mathfrak{g}$ (that is 0 on the other simple factors), so it is clear that $\mathfrak{g}_i$ must be $\mathbb{R}$-universal.

(2) See Lemma 2.3. (This means that all $\mathfrak{g}_i$ become inner over the same quadratic extension L.)

(3) We have

$$\begin{aligned} D_{\mathfrak{g},\mathbb{R}}(\lambda_i + \lambda_j) &\sim D_{\mathfrak{g},\mathbb{R}}(\lambda_i) \otimes_{\mathbb{R}} D_{\mathfrak{g},\mathbb{R}}(\lambda_j) \\ &\cong \big(D_{\mathfrak{g},\mathbb{Q}}(\lambda_i) \otimes_{\mathbb{Q}} \mathbb{R}\big) \otimes_{\mathbb{R}} \big(D_{\mathfrak{g},\mathbb{Q}}(\lambda_i) \otimes_{\mathbb{Q}} \mathbb{R}\big) \cong \mathbb{H} \otimes_{\mathbb{R}} \mathbb{H} \sim \mathbb{R}. \end{aligned}$$

Since $\mathfrak{g}$ is $\mathbb{R}$-universal, this implies $D_{\mathfrak{g},\mathbb{Q}}(\lambda_i + \lambda_j) \sim \mathbb{Q}$, from which we conclude that $D_{\mathfrak{g},\mathbb{Q}}(\lambda_i) \otimes_{\mathbb{Q}} D_{\mathfrak{g},\mathbb{Q}}(\lambda_j) \sim \mathbb{Q}$, so $D_{\mathfrak{g},\mathbb{Q}}(\lambda_i) \cong D_{\mathfrak{g},\mathbb{Q}}(\lambda_j)$ (since these are quaternion algebras).

(4) Let λ_j be a dominant weight of some (outer) simple factor G_j that is not fixed by the $*$-action. Then $\lambda_i + \lambda_j$ is not fixed by the $*$-action, so

$$D_{\mathfrak{g},\mathbb{Q}}(\lambda_i + \lambda_j) = D_{\mathfrak{g},L}(\lambda_i + \lambda_j) \sim D_{\mathfrak{g}_i,L}(\lambda_i) \otimes_L D_{\mathfrak{g}_j,L}(\lambda_j) = D_{\mathfrak{g}_i,L}(\lambda_i) \otimes_L L \cong D_{\mathfrak{g}_i,L}(\lambda_i).$$

Since $D_{\mathfrak{g},\mathbb{R}}(\lambda_i + \lambda_j) = D_{\mathfrak{g},\mathbb{C}}(\lambda_i + \lambda_j) = \mathbb{C}$, and $\mathfrak{g}$ is $\mathbb{R}$-universal, we conclude from Lemma 2.1(4) that $D_{\mathfrak{g}_i,L}(\lambda_i) = L$. Since $D_{\mathfrak{g},\mathbb{Q}}(\lambda_i) \otimes_{\mathbb{Q}} L \sim D_{\mathfrak{g},L}(\lambda_i)$, this means that $D_{\mathfrak{g},\mathbb{Q}}(\lambda_i)$ splits over L. □